Advanced Network Engineering Scenario-Based Examination

Background

The Uganda Industrial Research Institute operates two geographically separated campuses:

- **UIRI Namanve (LAN A)**
- **UIRI Nakawa (LAN B)**

Each campus has its own Local Area Network (LAN) consisting of one router, one Layer-3 core switch, and ten access switches. The two sites must communicate securely over a Wide Area Network (WAN) through a Site-to-Site VPN connection.

Management has tasked you, as the Network Engineer, with designing, implementing, configuring, securing, and documenting the entire network infrastructure to ensure reliable communication between the two campuses while providing Internet access and protecting organizational resources.

Existing Infrastructure

Site A – UIRI Namanve

The Namanve campus contains the following departments:

- Administration
- Finance
- Human Resource
- ICT
- Research
- CCTV/Surveillance

Available equipment:

- 1 Router
- 1 Layer-3 Core Switch
- 10 Access Switches
- DNS Server
- DHCP Server
- Web Server
- File Server

Site B – UIRI Nakawa

The Nakawa campus contains the following departments:

- Administration
- Finance
- Human Resource
- ICT
- Research
- CCTV/Surveillance

Available equipment:

- 1 Router
- 1 Layer-3 Core Switch
- 10 Access Switches
- DNS Server
- DHCP Server
- Database Server
- Email Server

Project Requirements

As the Network Engineer, you are required to perform the following tasks.

Task 1: Network Design

Design a complete physical and logical network topology for both campuses.

Your design should clearly show:

- Routers
- Core switches
- Access switches
- Servers
- End-user devices
- WAN connections
- VPN connectivity
- Internet connectivity

Task 2: IP Addressing and Subnetting

Develop a complete IP addressing scheme using VLSM.

Your addressing plan must:

- Support all departments at both campuses.
- Minimize address wastage.
- Provide room for future expansion.
- Include subnet masks, network addresses, broadcast addresses, and usable host ranges.

Task 3: VLAN Implementation

Create separate VLANs for:

- Administration
- Finance
- Human Resource
- ICT
- Research
- CCTV
- Network Management

Assign all switches and users appropriately.

Task 4: Inter-VLAN Routing

Configure inter-VLAN routing to enable communication between departments where permitted.

The solution should:

- Use the Layer-3 core switch.
- Provide gateway addresses for all VLANs.
- Ensure proper routing between VLANs.

Task 5: Dynamic Routing

Implement a dynamic routing protocol between the two campuses.

Requirements:

- Automatic route learning.
- Fast convergence.
- Scalability for future expansion.
- Exchange all internal routes between sites.

Task 6: WAN Connectivity

Configure the WAN connection between the two routers.

Your solution should:

- Provide communication between both campuses.
- Support dynamic routing.
- Allow VPN traffic to pass successfully.

Task 7: Site-to-Site VPN

Configure a secure VPN tunnel between the two campuses.

The VPN should:

- Encrypt all traffic between sites.
- Allow secure access to remote resources.
- Protect sensitive organizational information while traversing public networks.

Task 8: Network Address Translation (NAT)

Configure NAT to enable users at both campuses to access Internet services.

Your implementation should:

- Conserve public IP addresses.
- Support simultaneous access by multiple users.
- Allow internal users to browse the Internet.

Task 9: Switch Security

Implement security controls on all access switches.

Include measures to protect against:

- Unauthorized device connections
- Rogue DHCP servers
- MAC flooding attacks
- Spanning Tree attacks
- ARP spoofing
- Broadcast storms

Task 10: Router Security

Secure all routers by implementing:

- Secure remote management
- Password protection
- Login banners
- Access control policies
- Encrypted administrative access

Task 11: Monitoring and Management

Configure centralized network monitoring and management.

Your solution should support:

- Event logging
- Time synchronization
- Network monitoring
- Device management

Task 12: Redundancy and High Availability

Implement switching technologies that improve:

- Network reliability
- Loop prevention
- Link redundancy
- Load balancing

Task 13: Testing and Verification

Demonstrate and document that:

1. Devices within the same VLAN communicate successfully.
2. Inter-VLAN communication functions correctly.
3. The two campuses communicate successfully.
4. Routing information is exchanged dynamically.
5. Internet access is functional.
6. VPN connectivity is operational.
7. Security controls are working as expected.

Deliverables

Submit the following:

1. Physical Network Diagram.
2. Logical Network Diagram.
3. Complete IP Addressing Plan.
4. VLAN Design Table.
5. Dynamic Routing Configuration.
6. NAT Configuration.
7. VPN Configuration.
8. Switch Security Configuration.
9. Router Security Configuration.
10. Testing and Verification Results.
11. Comprehensive Network Documentation.

Examination Objective

Design, implement, configure, secure, troubleshoot, and document a multi-site enterprise network connecting **UIRI Namanve** and **UIRI Nakawa** through a secure WAN infrastructure while applying industry-standard routing, switching, VPN, NAT, and network security practices.